A FIELD GUIDE

Nine RAG Architectures, Compared.

From Standard to Multimodal — a practitioner's guide to when each architecture earns its place in production.

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01 Standard RAG
02 DeepRAG
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01 / 09 BEGINNER

Standard RAG

PIPELINE

01 User Query

02 Query Processing

03 Retrieval

04 Document Selection

05 Context Integration

06 LLM Response

★ TIP

Start with smaller chunk sizes (256–512 tokens) and experiment with overlap to improve coverage without overwhelming the context window.

▸ WHEN TO USE

Best for straightforward fact retrieval where accuracy matters but complexity is low.

◆ EXAMPLE

FAQ bots that surface up-to-date company policies or knowledge base entries.

02 / 09 INTERMEDIATE

DeepRAG

Hierarchical + Process Supervision

PIPELINE

01 User Query

02 Question Decomposition

03 Retrieval per sub-question

04 Reasoning + Verification

05 Final Response

★ TIP

Use hierarchical indicators to manage nested dependencies, and combine process supervision (reward signals) to align sub-queries with final answers.

▸ WHEN TO USE

Multi-hop or domain-specific (e.g. biomedical) questions needing structured decomposition.

◆ EXAMPLE

Biomedical QA over linked concept databases (e.g. UMLS).



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03 / 09 INTERMEDIATE

MA-RAG

Multi-Agent RAG

PIPELINE

01 User Query

02 Planner + Extractor + Retriever

03 Combined Retrieval & Reasoning

04 LLM Output

★ TIP

Each agent handles a subtask, and only the necessary agents are invoked per query. Agents use chain-of-thought prompting for interpretability.

▸ WHEN TO USE

Ambiguous or complex queries where decomposition and agent collaboration helps.

◆ EXAMPLE

Open-domain QA across heterogeneous sources.



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ADVANCED

Corrective RAG

PIPELINE

01 User Query

02 Initial Response

03 Error Detection

04 Retrieval

05 Response Correction

06 Final Response

★ TIP

Use a lightweight verification loop — either a separate classifier or an auxiliary LLM — to flag uncertain segments and trigger corrective retrieval.

▸ WHEN TO USE

Suitable when outputs must be double-checked before being shown to the user.

◆ EXAMPLE

Clinical decision-support assistants that cross-verify against trusted medical literature.

05 / 09 INTERMEDIATE

Speculative RAG

PIPELINE

01 User Query

02 Small Model Draft

03 Parallel Retrievals

04 Large Model Verification

05 Final Response

★ TIP

Pair a smaller, faster model for drafting with a larger model that validates against retrieved documents; adjust retrieval parallelism based on latency budgets.

▸ WHEN TO USE

Ideal when response speed is crucial but correctness can't be sacrificed.

◆ EXAMPLE

Live chat agents for e-commerce that need rapid draft answers while verifying product details.



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06 / 09 ADVANCED

Fusion RAG

PIPELINE

01 User Query

02 Multiple Query Generation

03 Multi-Retrieval

04 Fusion of Documents

05 LLM Response

★ TIP

Employ a fusion method like Reciprocal Rank Fusion to merge results, plus weighting sources according to reliability and domain relevance.

▸ WHEN TO USE

Useful when queries require pulling evidence from different types of sources or formats.

◆ EXAMPLE

Academic research assistants synthesizing across journal articles, patents, and technical blogs.



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07 / 09 ADVANCED

RAG-Gym

Process-Supervised Agentic RAG

PIPELINE

01 User Query

02 Agentic Planning

03 Step-by-step Retrieval

04 LLM Generation + Supervision

★ TIP

Use process-level rewards to train agents — reward which retrieval or reformulation step is better, not just the final answer. The RAG-Gym / ReSearch architecture is from a recent arXiv paper.

▸ WHEN TO USE

For multi-step reasoning or tasks that benefit from supervising intermediate steps.

◆ EXAMPLE

QA systems that have to plan a search path rather than one-shot retrieval.

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08 / 09 ADVANCED

Modular RAG

PIPELINE

01 User Query

02 Module Routing

03 Retrieval / Rewriting / Fusion

04 LLM Generation

★ TIP

Decompose into operators (routing, query rewriting, fusion) so you can replace your reranker or reformulator independently. The Modular RAG paper identifies linear, conditional, branching, and looping patterns.

▸ WHEN TO USE

When you want flexibility in customizing or swapping parts of the RAG pipeline.

◆ EXAMPLE

Research systems exploring new retrieval/fusion strategies.

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09 / 09 ADVANCED

SAM-RAG

Self-adaptive Multimodal RAG

PIPELINE

01 Multi-modal Query

02 Adaptive Cross-Modal Retrieval

03 Filtering & Quality Check

04 LLM Response

★ TIP

Dynamically select which modalities to retrieve from, verify the relevance of retrieved multimodal data, and integrate them in context. The model adapts retrieval count and modality per query.

▸ WHEN TO USE

When queries or data involve images, audio, or mixed modalities.

◆ EXAMPLE

Q&A combining text + images (e.g. product descriptions + photos).

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